

Homework 7

40 points

1. Given, $f(x) = (x^2 + 2x + 3)^5$. Find the first and second order derivative of the given function. **3 + 3 = 6 points**

$$\begin{aligned} \bullet \quad f'(x) &= 5 (x^2 + 2x + 3)^4 (2x + 2) \\ &= 10 (x^2 + 2x + 3)^4 (x + 1) \end{aligned}$$

$$\begin{aligned} \bullet \quad f''(x) &= 10 [(x^2 + 2x + 3)^4 + 4 (x^2 + 2x + 3)^3 \\ &\quad (2x + 2)(x + 1)] \\ &= 10 (x^2 + 2x + 3)^3 [(x^2 + 2x + 3) + \\ &\quad 8 (x^2 + 2x + 1)] \\ &= 10 (x^2 + 2x + 3)^3 [x^2 + 2x + 3 + 8x^2 \\ &\quad + 16x + 8] \\ &= 10 (x^2 + 2x + 3)^3 (9x^2 + 18x + 11) \end{aligned}$$

2. Given $y = \sin(x^2 + 3x)$. Find $\frac{dy}{dx}$ **4 points**

$$\frac{dy}{dx} = \cos(x^2 + 3x)(2x + 3)$$

3. Find all values of x for which the tangent to $f(x) = x^4 - 4x^3 + 3$ will be horizontal. **5 points**

- horizontal tangent @ $f'(x) = 0$

- $f'(x) = 4x^3 - 12x^2$
 $= 4x^2(x - 3)$

$$f'(x) = 0 \text{ when } 4x^2 = 0 \text{ or } x - 3 = 0$$
$$\hookrightarrow x = 0 \qquad \qquad \qquad \hookrightarrow x = 3$$

4. Find f_x and f_y for the following: **5*2 = 10 points**

$$\frac{dn' - nd'}{d^2}$$

- $f(x, y) = x^2y + 3xy^2$
- $f(x, y) = \frac{xy}{x^2+y^2}$ (for $(x, y) \neq (0, 0)$)

$$\bullet f_x(x, y) = 2xy + 3y^2$$

$$\bullet f_y(x, y) = x^2 + 6xy$$

$$\begin{aligned}\bullet f_{xx}(x, y) &= \frac{(x^2 + y^2) \cdot y - (xy)(2x)}{(x^2 + y^2)^2} \\ &= \frac{y(x^2 + y^2 - 2x^2)}{(x^2 + y^2)^2} = \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}\end{aligned}$$

$$\begin{aligned}\bullet f_{yy}(x, y) &= \frac{(x^2 + y^2) \cdot x - (xy)(2y)}{(x^2 + y^2)^2} \\ &= \frac{x(x^2 + y^2 - 2y^2)}{(x^2 + y^2)^2} = \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}\end{aligned}$$

5. Compute all second order partial derivatives of $f(x, y) = x^3y^2 - 5x^2y$ 4*2
= 8 points

1st order

$$\bullet f_x(x, y) = 3x^2y^2 - 10xy$$

$$\bullet f_y(x, y) = 2x^3y - 5x^2$$

2nd order

$$\bullet f_{xx}(x, y) = 6xy^2 - 10y$$

$$\bullet f_{xy}(x, y) = 6x^2y - 10x$$

$$\bullet f_{yx}(x, y) = 6x^2y - 10x$$

$$\bullet f_{yy}(x, y) = 2x^3$$

> same!

6. Given $f(x, y, z) = x^2 + y^2 + z^2$

- Find the gradient vector at point $(1, 2, 2)$ **3 points**
- Find the directional derivative at that point in the direction of the vector $(1, 1, 1)$ **2 points**
- Find the maximum rate of change. **2 points**

$$\nabla f(x, y, z) = (2x + 2y + 2z)$$

$$\nabla f(1, 2, 2) = (1, 2, 2)$$

$$\text{Unit vector: } \hat{v} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$\text{Directional deriv.: } D_{\hat{v}} f = \nabla f \cdot \hat{v}$$

$$= (2, 4, 4) \cdot \frac{1}{\sqrt{3}} (1, 1, 1) = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3}$$

$$\text{Max. rate of change: } |\nabla f| = \sqrt{2^2 + 4^2 + 2^2} = 6$$